University of Florida, Department of Mathematics

Analysis First-Year Exam, August 2025, Part 1

Answer FOUR questions in detail. State carefully any results used without proof.

1. Let $A \subseteq \mathbb{R}$ be closed under subtraction, in the sense that if $s, t \in A$ then $s - t \in A$. Prove that if $A \cap (0, u) \neq \varnothing$ whenever $u > 0$ then A is dense in $\mathbb{R}$.

2. For every positive integer n let A_n be a subset of the metric space X. For each of the following statements, give a proof or a counterexample. Here, an overline signifies closure.

(i) if N is a positive integer then $\overline{\bigcup_{n=1}^{N} A_n} = \bigcup_{n=1}^{N} \overline{A_n}$;

(ii) $\overline{\bigcup_{n=1}^{\infty} A_n} = \bigcup_{n=1}^{\infty} \overline{A_n}$.

3. Let S be a countably infinite subset of $\mathbb{R}^2$ (with the Euclidean metric). Must the complement $\mathbb{R}^2 \setminus S$ be connected? If so, give a proof; if no, give a counterexample.

4. Let the nonempty closed subsets A and B of a metric space be disjoint. Must the infimum $\inf\{d(a, b) : a \in A, b \in B\}$ be strictly positive? Does the answer change if either A or B is compact?

5. Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable, with $f'(0) = 1$. Must there exist an open interval about 0 on which f is increasing? Proof or counterexample.